for space debris removal by bi-directional momentum ejection from a satellite. When plasmas carrying momentum fluxes $F_1$ and $F_2$ are expelled from two axially opposite satellite exits, the respective forces shown by the horizontal arrows $F_1$ (pointing to the left and providing the acceleration of the satellite with respect to the orbit velocity) and $F_2$ (providing the deceleration) are generated and used to adjust the satellite velocity relative to the debris. Continuously imparting momentum flux $F_1$ to the debris (horizontal arrow $F_1$ pointing to the right) will cause its deceleration, final re-entry into the Earth atmosphere and natural burn up.

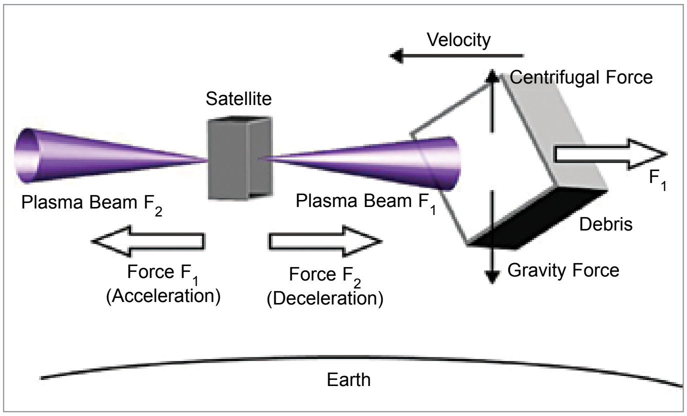


*Fig. 4: Space debris removal by bi-directional momentum ejection from a satellite*

***Recycling satellites.*** Instead of just trashing space debris, some dead satellites could be 'mined' by other satellites for useable components. DARPA's Phoenix plan could create new technology to enable harvesting of some valuable components from satellites in so-called 'graveyard' orbits. The plan would work to devise nano-satellites that would be cheaper to launch and could essentially complete their own construction by latching onto an existing satellite in the graveyard orbit and using the parts it needs.

***Space garbage trucks.*** The US Defense Advanced Research Project Agency (DARPA) is investing in the Electrodynamic Debris Eliminator, or EDDE, a space 'garbage truck' equipped with 200 giant nets, which could be extended out to scoop up space garbage. The EDDE could then either fling the garbage back to Earth to land in the oceans, or push the objects into a closer orbit, which would keep them out of the way of current satellites until they decay and fall back to Earth.

***Self-destructing janitor satellites.*** Swiss researchers at the Federal Institute of Technology have devised a small satellite, called CleanSpace One, which could find and then grab onto space junk with jellyfish-like tentacles. The device would then plummet back towards Earth, where both the satellite and the space debris would be destroyed by the heat and friction of re-entry.

***Giant lasers.*** Using high-powered pulsed lasers based on Earth to focus plasma jets on space debris could cause them to slow down slightly and to then re-enter and either burn up in the atmosphere or fall into the oceans. "The method is called Laser Orbital Debris Removal (LODR) and it wouldn't require new technology to be developed—it would use laser technology that has been around for 15 years. It would be relatively cheap, and readily available." The biggest hitch, other than adding more litter to the oceans, is the estimated $1 million per object price tag.

***Space balloons.*** The Gossamer Orbit Lowering Device, or GOLD system, uses an ultra-thin balloon (thinner than a plastic sandwich bag), which is inflated with gas to the size of a football field and then attached to large pieces of space debris. The GOLD balloon will increase the drag of objects enough so that the space junk will enter the earth's atmosphere and burn up. If the system works, it could speed up the re-entry of some objects from a couple of hundred years to just a few months.

***Wall of water.*** Another idea for cleaning up space junk, from James Hollopeter of GIT Satellite, is to launch rockets full of water into space. The rockets would release their payload to create a wall of water that orbiting junk would bump into, slow down, and fall out of orbit. The Ballistic Orbital Removal System is said to be able to be put into action inexpensively, by launching water on decommissioned missiles.

***Space pods.*** Russia's space corporation, Energia, is planning to build a space pod to knock junk out of orbit and back down to Earth. The pod is said to use a nuclear power core to keep it fueled for about 15 years as it orbits the earth, knocking defunct satellites out of orbit. The debris would either burn up in the atmosphere or drop into the ocean. It is claimed to clean up the space around Earth in just ten years, by collecting around 600 dead satellites (all on the same geosynchronous orbit) and them sinking them into the ocean.

***Sticky booms.*** Altius Space Machines is currently developing a robotic arm system it calls a 'sticky boom,' which can extend up to 100 metres, and uses electro-adhesion to induce electrostatic charges onto any material (metal, plastics, glass, even asteroids) it comes into contact with, and then clamp onto the object because of the difference in charges. The sticky boom can attach to any space object, even if it was not designed to be handled by a robotic arm. The sticky boom could be used to latch onto space debris for disposal. EFY